

SIMATS Engineering

SIMATS Online Programming Lab — Python Programming (Class Test)

Course Name	Python Programming
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Topic	CO3-AT2-CLASS TEST
Question	Q2 — Min, Max, and Average of a List using Tuples (1 Mark)

Q2. [1 Mark]

Write a Python function that accepts a list of numbers and returns a tuple containing the minimum, maximum, and average values. Use tuple unpacking to store and print the returned values.

Approach / Design

- **Function** (get_min_max_avg): takes a list of numbers and computes the minimum with min(), the maximum with max(), and the average by dividing the sum by the count, then returns all three as a single tuple (minimum, maximum, average).
- **Tuple unpacking**: the caller unpacks the returned tuple directly into three separate variables (lowest, highest, avg) in one line, avoiding manual indexing into the tuple.
- **Display**: each unpacked value is printed with a descriptive label.

Python Program

```
"""
CO3-AT2 - Class Test
Q2: Min, Max, and Average of a List using Tuples [1 Mark]
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Defines a function that accepts a list of numbers and returns a
tuple containing the minimum, maximum, and average values. Tuple
unpacking is used to store and print the returned values.
"""

def get_min_max_avg(numbers):
    """
    Returns a tuple (minimum, maximum, average) for the given
    list of numbers.
    """
    minimum = min(numbers)
    maximum = max(numbers)
    average = sum(numbers) / len(numbers)
    return (minimum, maximum, average)

def main():
    numbers = [30, 32, 31, 29, 35, 33, 30]

    # Tuple unpacking to store the returned values
    lowest, highest, avg = get_min_max_avg(numbers)

    print("Minimum:", lowest)
    print("Maximum:", highest)
    print("Average:", avg)
```

```
if __name__ == "__main__":  
    main()
```

Sample Output

```
Minimum: 29  
Maximum: 35  
Average: 31.428571428571427
```

Conclusion

The function `get_min_max_avg()` bundles the minimum, maximum, and average of a numeric list into a single tuple, which the caller unpacks into three named variables in one step. For the sample list `[30, 32, 31, 29, 35, 33, 30]`, this correctly returns a minimum of 29, a maximum of 35, and an average of approximately 31.43, demonstrating both multi-value returns and tuple unpacking in a single, reusable function.